

REPORT

Title

Prolog Program to Determine Whether a Particular Bird Can Fly or Not

Aim

To write a Prolog program to determine whether a given bird **can fly or cannot fly** using facts and rules.

Problem Statement

Create a Prolog database containing different birds and their flying abilities. The program should accept a bird name as a query and display whether the particular bird can fly or not.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp6.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp6.pl`.
3. Enter the bird facts and rules.
4. Save the program.
5. Load the file using `[exp6] ..`
6. Query a particular bird.
7. Observe whether the bird can fly or not.

Program

```
bird(sparrow).  
bird(eagle).  
bird(parrot).  
bird(pigeon).
```

```
bird(penguin).
bird(ostrich).

can_fly(sparrow).
can_fly(eagle).
can_fly(parrot).
can_fly(pigeon).

cannot_fly(penguin).
cannot_fly(ostrich).

fly(Bird) :-
    can_fly(Bird).

fly(Bird) :-
    cannot_fly(Bird),
    fail.
```

Sample Output

Query:

```
?- [exp6].
true.

?- can_fly(eagle).
true.
```

For a bird that cannot fly:

```
?- cannot_fly(penguin).
true.
```

Another query:

```
?- can_fly(penguin).
false.
```

Conclusion

Thus, the Prolog program was successfully implemented to determine whether a **particular bird can fly or not**. The program correctly identifies flying birds such as **eagle and sparrow** and non-flying birds such as **penguin and ostrich**.